

When Intelligent HVAC Control Claims Savings, What Makes the Evidence Admissible?

Independent Technical Governance Reference | Version 2.0 | 8 October 2026

PRE-EXAMINATION - BASELINE NOT FROZEN - NO RESEARCHER OR INSTITUTIONAL ENDORSEMENT

No formal ALLOW, HOLD, DENY, or ESCALATE disposition is issued by this document.

Research examined: Khabbazi, A. J., et al., *Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review*, arXiv:2503.05022v6 (19 July 2025).

Purpose: Translate source-reported experimental reliability lessons into a bounded TA14 examination framework for a future, specifically identified building-level claim or controller intervention.

Important separation: The research authors assess field-test evidence and performance reporting. TA14 separately asks whether a proposed consequence has admissible evidence, applicable authority, and established standing to become reality NOW.

01 | Executive examination summary

Advanced HVAC control has credible potential, but a published performance result is not an automatic authorization to change a building. The reviewed literature contains 104 papers and 206 field tests. The authors found that 71% of reviewed papers used experiment protocols that may produce unreliable performance estimates. This is **not** a finding that 71% of deployed controllers failed. [S1, Abstract; Sections 4-5]

Source-reported result	Review-level figure
Residential / commercial papers	24 / 80
Residential / commercial field tests	52 / 154
Papers with potentially unreliable protocols	71%
Filtered duration-weighted energy savings	22% residential; 15% commercial
Filtered duration-weighted utility-cost savings	16% residential; 13% commercial
Papers addressing deployment / operating / maintenance costs	13 of 104

These values summarize reviewed experiments. They do not establish a forecast, certification, warranty, or authority for any individual building. [S1, Abstract; graphical abstract; Sections 5-6]

TA14 provisional assessment: The review is suitable for an independent, source-grounded pre-examination. No identified claimant, frozen building baseline, verified authority chain, or proposed live control action has been submitted for adjudication.

02 | Examination scope, method and exclusions

Examination object

The object is the **review paper and its stated methodological findings**, not the original sensor streams, building operations, or individual controllers in the 104 underlying studies.

Method used for this reference

- Read the supplied review and distinguish authors' reported results from TA14 interpretation.
- Identify protocol issues: benchmark selection, thermal boundary, test duration, operating context and implementation costs.
- Map those issues to the TA14 chain: Reality, Record, Continuity, Admissibility, Binding, Commit, Execution, Outcome.
- Define the additional evidence required before a specific building intervention could be examined.
- Use provisional evidence-gap descriptions rather than formal dispositions in the absence of an actual claim.

Not performed

- No independent replication of statistical calculations or reanalysis of the open-access study dataset.
- No inspection of original building meters, controller logs, calibration certificates or occupant complaints.
- No independent assessment of legal authority, contracts, insurance, licensure or owner standing for a real asset.
- No endorsement, certification, commissioning approval, or control authorization.

A source-based interpretation cannot silently substitute for an independent field audit. The boundary between those two activities remains explicit throughout this reference.

03 | Evidence provenance and claim classes

Claim class	Permitted use in this reference
Source-reported	Attribute to the review; include section / figure reference.
TA14 interpretation	Identify as a governance inference or examination question, not an author conclusion.
Illustrative scenario	Label hypothetical; no simulated values may be passed off as measured evidence.
Building-specific determination	Not available until asset, claimant, frozen baseline, evidence packet and standing are examined.

Core provenance rules

Preserve the distinction between a paper count and a test count; between energy savings and cost savings; between a duration-weighted review aggregate and an individual building result; and between measured baselines and simulated counterfactuals. [S1, Sections 3-5]

The authors' filtered results are based on a subset of the reviewed evidence, not all 206 tests. The source describes 25 residential and 28 commercial tests meeting the protocol filter; the available savings metrics involve smaller subsets, depending on the reported outcome. [S1, Section 5; Figure 10]

Evidence ledger fields

Every future claim should have a claim ID, source or instrument ID, source date, responsible custodian, immutable original or verified copy, transformation record, uncertainty statement, applicable asset, context window, and reviewer disposition. These are **proposed TA14 packet requirements**, not requirements asserted by the research authors.

04 | The three experimental reliability boundaries

Boundary	Research recommendation	TA14 evidence consequence
Measured comparison	Compare against a measured baseline controller in the same or similar nearby building.	Identify counterfactual, adjustment method, and original meter records.
Physical scope	Use a whole building, an insulated subset, or account for heat transfer across zones.	Preserve zone map, load transfers and claimed measurement boundary.
Duration	At least 7 days; ideally 1 month or more.	Preserve start/end, operational exceptions and representativeness.

These are the **authors' recommendations** for trustworthy field demonstrations. They are not universal legal thresholds or automatic TA14 pass criteria. [S1, Section 6.3.1]

Why this matters

A short trial can mistake thermal storage for persistent savings. A partial-zone trial can transfer demand to adjacent areas. A simulation-based comparison can be useful research but cannot be treated as a measured operational baseline without disclosure. [S1, Section 5]

Admissibility is claim-specific. Meeting three screening checks may improve the performance estimate while leaving identity, continuity, authorization, equipment safeguards and actual outcomes unexamined.

05 | Context, economics and generalizability

Operating context

The review identifies inconsistent reporting of tariff structures, building envelope and thermal properties, climate and control architecture. These factors limit rigorous comparisons between studies. [S1, Section 6.1; 6.3.3]

Economic context

Only 13 of the 104 reviewed papers reported deployment, operation or maintenance cost information or related challenges. The review distinguishes one-time development work from repeated site-level costs. [S1, Sections 6.2-6.3]

Cost class	Illustrative documentation required
Hardware / sensing	Meters, sensors, gateways, actuators, installation, calibration.
Integration / IT	Data access, cybersecurity approvals, networking, API and subscription costs.
Engineering labor	Model development, commissioning, tuning, troubleshooting and handoff.
Ongoing operations	Monitoring, recalibration, repairs, subscriptions and service visits.
Financial analysis	Gross and net savings, tariff, baseline adjustments, uncertainty, life-cycle costs.

The review includes examples of 190 engineer-days for one residential deployment and 239 person-days for a commercial deployment; these are examples from particular studies, not representative market prices. [S1, Section 6.2.3]

06 | TA14 eight-anchor consequence chain

Anchor	Examination question
01 Reality	What physically occurred in the operating building?
02 Record	Were raw measurements, timestamps, setpoints and controller states preserved?
03 Continuity	Can records and transformations be traced without unexplained gaps?
04 Admissibility	Is evidence sufficient for this precise claim, including uncertainty?
05 Binding	Which asset, control envelope, time window and consequence does it support?
06 Commit	Who has standing and applicable authority to approve the action now?
07 Execution	Was the authorized command executed within scope and safeguards?
08 Outcome	Were savings, comfort, safety and unintended consequences verified?

This chain is a **TA14 governance overlay**. It is not presented as a methodology used or endorsed by the review authors.

The decision boundary

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

A controller recommendation is a proposed action. A valid performance estimate is evidence about a claim. Neither alone constitutes permission to actuate equipment.

07 | Worked example: a proposed HVAC control change

Illustrative scenario only: A model-predictive controller recommends increasing an office cooling setpoint during an afternoon demand peak. Its forecast predicts lower electricity cost while keeping occupied-zone temperatures within configured limits. No actual building, measurements or savings are represented here.

Gate	Required examination
REALITY / RECORD	Current temperatures, occupancy, equipment mode, actual demand and timestamped measurements.
CONTINUITY / ADMISSIBILITY	Meter quality, baseline method, forecast uncertainty, missing data and thermal effects on adjacent zones.
BINDING	Specific AHU, zones, command limits, permitted schedule, expiration and affected occupants.
COMMIT	Documented owner or delegated operator approval; standing and approval validity.
EXECUTION	Authenticated command, constraint checks, override availability and auditable command receipt.
OUTCOME	Actual demand and energy, occupant comfort, cycling, complaints and rebound demand.

Example stop conditions

- A sensor becomes stale or unreliable.
- A zone crosses an approved comfort or safety boundary.
- The approval expires or the affected asset changes.
- A maintenance override, emergency state or material occupancy change occurs.

The proposed TA14 rule is to stop or revalidate the decision when relevant context changes. This example illustrates a governance design, not a claim that the review tested this approval workflow.

08 | Provisional issue register - not formal dispositions

The entries below identify what is **missing for a future building-level examination**. They are not findings against the paper, its authors, or every underlying experiment.

Issue	Pre-examination state	Closure evidence
Measured comparability	Evidence not submitted	Measured baseline, adjustments, weather, occupancy, uncertainty.
Thermal boundary	Evidence not submitted	Whole-building metering or documented interzone transfers.
Continuity	Evidence not submitted	Time series, exceptions, version history, missing-data log.
Net economics	Evidence not submitted	Capital, engineering, operating and maintenance ledger.
Authority / standing	Not examined	Owner/delegate credentials, scope, validity and consent.
Verified consequence	Not examined	Post-command records, comfort, safety and cost reconciliation.

Disposition discipline

ALLOW / HOLD / DENY / ESCALATE should be reserved for a defined proposed consequence evaluated under a documented examination. A missing packet is not automatically a formal HOLD. When no applicant, asset or action is before TA14, the accurate statement is **not examined** or **evidence not submitted**.

09 | Pilot evidence packet: identity and observation

Packet A - identity, scope and standing

- Building and asset identifiers; owner and delegated operator identities; roles and evidence of standing.
- Controller vendor, software/model version, change history and control point inventory.
- Controlled zones, adjacent zones, occupancy classes, mechanical systems and known safety limits.
- Precisely worded proposed consequence and requested decision validity window.

Packet B - observation and preserved record

- Raw meter readings, sensor records, timestamp convention and clock synchronization.
- Instrument inventory, calibration status, sampling frequency and known measurement errors.
- Baseline controller schedule and setpoints; actual test controller commands and acknowledgments.
- Weather, occupancy, tariff, maintenance events, overrides and missing-data register.

Baseline freeze checklist

- Identify the asset, measurement boundary, baseline dates and control configuration.
- Preserve raw observations before normalization or adjustment.
- Record approved analysis method, uncertainty and exclusions.
- Assign custodian, version, freeze timestamp and change-control process.

No baseline has been frozen for this reference. The checklist is a proposed requirement for a future pilot, not a statement about the reviewed studies.

10 | Pilot evidence packet: analysis and authorization

Packet C - comparability and analytical record

- Baseline selection, measured comparator and reasons for exclusions.
- Weather and occupancy normalization method, model assumptions and statistical uncertainty.
- Whole-building or subset treatment; explicit adjacent-zone heat-transfer accounting.
- Raw and adjusted results stored separately, with reproducible calculation scripts or procedures.
- Energy, demand and utility cost distinguished; tariffs and billing periods preserved.

Packet D - authority and commit

- Source of applicable authority: contract, owner delegation, operational policy or other documented basis.
- Standing of the person or system requesting, approving and executing the consequence.
- Specific assets, command types, operating bounds, validity period and emergency overrides.
- Conditions for revalidation, escalation and withdrawal of approval.
- Human review and audit trail where required by the applicable authority framework.

Proposed evidence-to-decision record

Field	Example content (not real data)
Claim / action ID	Pilot-XX / Setpoint-change-XX
Evidence snapshot	Immutable dataset version and analysis hash
Authority source	Owner delegation reference and scope
Decision / validity	Not issued; time-bounded if later approved
Revalidation trigger	Sensor, occupancy, equipment or policy change

11 | Execution, outcomes and changed context

Packet E - execution

Record the exact proposed command, authorizing decision, command dispatch and acknowledgement, execution timestamp, resulting equipment state, safety interlocks, manual overrides, and deviations. A proposal that was never executed cannot be reported as an executed consequence.

Packet F - outcome

Verify realized energy and cost effects against the approved comparison method, and also document occupant comfort, humidity or other indoor environmental conditions, equipment cycling, alarms, complaints, rebound load, operating labor and maintenance cost. Preserve null and adverse outcomes rather than reporting savings alone.

Changed-context revalidation

Trigger	Required response concept
Material sensor or data failure	Suspend unsupported automated action; inspect evidence.
Changed occupancy or building use	Reassess scope, comfort constraints and validity.
Equipment or control-system replacement	Rebind evidence and approval to the new configuration.
Approval expiration / authority change	Do not assume continuing permission.
Safety or emergency condition	Apply approved protective override and preserve the event.

These are proposed TA14 governance controls. They are not empirically tested outcomes reported by Khabbazi et al.

12 | Reviewer protocol and escalation boundaries

Review sequence

- Verify that the applicant and the requested consequence are specifically identified.
- Confirm standing and applicable authority for the asset and action.
- Inventory evidence and test provenance, continuity, relevance and uncertainty.
- Separate observed values, inferred values and requested permissions.
- Document gaps and request bounded remediation where appropriate.
- Only then consider a formal, time- and scope-bounded disposition under the governing TA14 process.

Decision semantics

Label	Use only after formal examination
ALLOW	The specified consequence meets applicable criteria within stated bounds.
HOLD	The defined decision cannot proceed pending identified remediable conditions.
DENY	The defined consequence fails a governing requirement.
ESCALATE	A higher or different authority must resolve the decision.

This reference issues none of these dispositions. A research paper does not supply the asset-specific standing, authority and live evidence needed for an execution decision.

Observed is not inferred. Inferred is not authorized.

A favorable performance estimate must never be silently promoted into execution permission.

13 | Eight-image teaching and showroom specification

Image	Anchor	Visual lesson
01	Reality	Actual HVAC operation vs forecast / simulation; show physical conditions.
02	Record	Meter, sensor and command records preserved with timestamps.
03	Continuity	Trace baseline, test period, controller version and exceptions.
04	Admissibility	Measured benchmark, physical scope and adequate duration.
05	Binding	Attach evidence to a specific asset, claim and permitted control envelope.
06	Commit	Standing, delegated authority and bounded approval.
07	Execution	Authenticated command path, safeguards and stop/revalidate trigger.
08	Outcome	Measured energy and cost plus comfort and equipment consequences.

Production rules

Produce images in numeric order, one image per step. Label them **illustrative teaching graphics**, not original study data. Use this PDF as the narrative source. Samantha narration follows each number; do not embed Samantha as a person in the images. Preserve free user scrolling and keep audio controls from obscuring the graphics.

Showroom controls

Display the independent-pre-examination status prominently. Provide a working PDF download, source citation, numbered visual navigation, accessible narration controls, and a statement that no researcher, university, laboratory or manufacturer endorsement is implied.

14 | Source notes and verification boundaries

[S1] Primary source: Khabbazi, Arash J., Elias N. Pergantis, Levi D. Reyes Premer, Panagiotis Papageorgiou, Alex H. Lee, James E. Braun, Gregor P. Henze, and Kevin J. Kircher. *Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review*. arXiv:2503.05022v6, 19 July 2025. <https://arxiv.org/abs/2503.05022>

Source location	Used for
Abstract / graphical abstract	Paper counts, test counts, 71% finding, filtered savings and cost reporting.
Section 4	Study demographics, control implementations and test protocols.
Section 5 / Figure 10	Experimental reliability, filtering and reported savings.
Section 6.1	Building and tariff context; limits of generalization.
Sections 6.2-6.3	Deployment costs, research recommendations and protocol guidance.
Appendix B	Study-level summaries, not independently verified in this reference.

Source limitation

The cited review and its appendices provide published descriptions and aggregated findings. This reference did not inspect or independently reproduce the authors' underlying open-access dataset, original measurements, or field controller records. Thus its conclusions are interpretive and provisional.

Independent status

TA14 is not affiliated with, endorsed by, or acting on behalf of the research authors, Purdue University, University of Colorado Boulder, NREL, Trane Technologies, or any other institution named in the review.

15 | Publication readiness and version control

What is ready

- A bounded, research-attributed technical interpretation.
- A TA14 examination method and eight-anchor teaching framework.
- An illustrative proposed consequence with evidence and authority gates.
- A practical pilot evidence packet and pre-examination issue register.

What is not yet established

- A real applicant, identified asset and proposed live action.
- A frozen measured baseline and verified instrument records.
- A validated authorization and standing chain.
- A formal disposition, certified savings result or independently verified pilot outcome.

Version record

v1.0: Eight-page initial technical reference. **v2.0:** Expanded method, source traceability, worked decision example, evidence packet, disposition discipline and production specification. Date: 8 October 2026.

Publication status: Suitable as an independent technical pre-examination reference after editorial approval. It is not a formal examination ruling or operational control authorization.

Governing question: Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?